

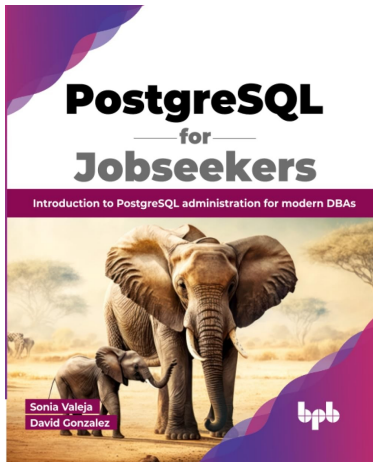
Database Administration

Lecture 06: Backup and Restore.

Valeja & Gonzales.

14 de septiembre de 2026

Database Administration: Backup and Restore.



Content has been extracted from *PostgreSQL for Jobseekers* (Chapter 7), by Sonia Valeja and David Gonzales, 2023. Visit <https://bpbonline.com/products/postgresql-for-jobseekers>.

Plan

Introduction

Types of Backups

- Physical

- Logical

Types of Restore

Advanced Backup Tools

Introduction

- ▶ Regular backups are crucial for database administration.
- ▶ Backups enable recovery from accidental changes.
- ▶ PostgreSQL provides multiple backup and restore tools.

Plan

Introduction

Types of Backups

Physical

Logical

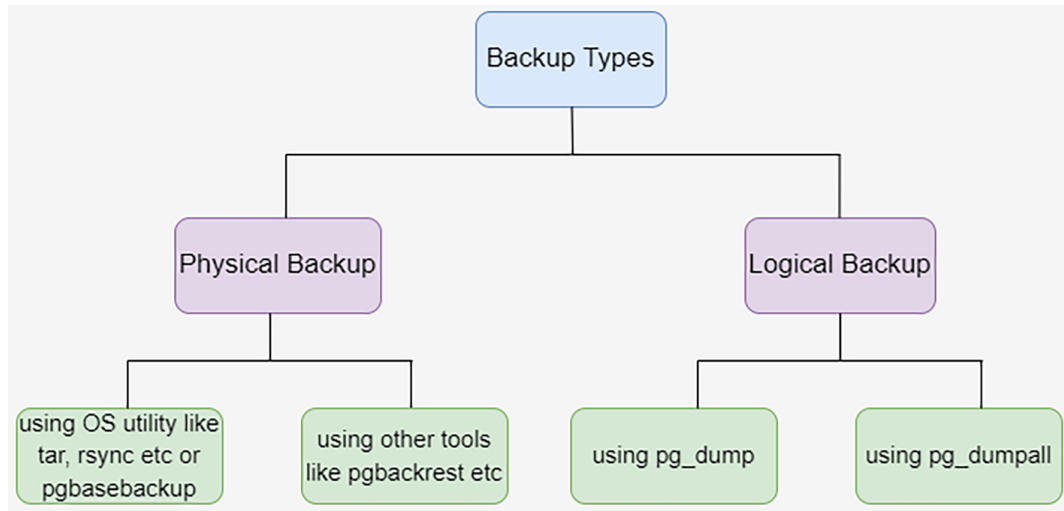
Types of Restore

Advanced Backup Tools

Types of Backups

- ▶ **Physical Backup:** Copies entire data directory.
- ▶ **Logical Backup:** Exports data in human-readable format.

Types of Backups



Physical Backup

- ▶ Uses filesystem-level copying.
- ▶ Steps:
 1. Create a checkpoint.
 2. Copy data directory (e.g., using `tar` or `rsync`).
 3. Stop backup process.
- ▶ Example tool: `pg_basebackup`.

pg_basebackup

```
postgres@ip-172-31-27-8:~$ pg_basebackup --help
```

pg_basebackup takes a base backup of a running PostgreSQL server.

Usage:

pg_basebackup [OPTION]...

Options controlling the output:

- D, --pgdata=DIRECTORY receive base backup into directory
- F, --format=plt output format (plain (default), tar)
- r, --max-rate=RATE maximum transfer rate to transfer data directory
(in kB/s, or use suffix "k" or "M")
- R, --write-recovery-conf
write configuration for replication
- T, --tablespace-mapping=OLDDIR=NEWDIR
relocate tablespace in OLDDIR to NEWDIR
- waldir=WALDIR location for the write-ahead log directory
- X, --wal-method=none|fetch|stream
include required WAL files with specified method

Point in Time Recovery (PITR) / Archival

PITR by enabling **archiving**, this allows restoring the database to a specific time frame, provided that WAL¹/archives are restored until that point.

Steps:

- ▶ Enable archiving in `postgresql.conf`.
- ▶ Configure the following parameters:

Configuration Example:

```
archive_mode = on
archive_command = 'test ! -f /mnt/server/archivedir/%f && \
    cp %p /mnt/server/archivedir/%f' # Linux
archive_command = 'copy "%p" "C:\\server\\archivedir\\%f"' # Windows
```

We will see in the restore section how to recover a database to a specific time frame.

¹Write-Ahead Logging

Pros and Cons of Physical Backup

Pros:

- ▶ Contains a full filesystem-level backup of the data directory.
- ▶ Includes configuration files such as `postgresql.conf`, `pg_hba.conf`, etc.
- ▶ Simple process: copy the data directory and custom tablespaces to the destination server and start PostgreSQL.

Cons:

- ▶ Only cluster-level backups are possible (no single database or table backup).
- ▶ Consistency of the backup is difficult to guarantee.
- ▶ Backup/restore speed depends heavily on server hardware and network speed (for remote copies).

Logical Backup

- ▶ Exports data in SQL format.
- ▶ Allows individual table or database backups.
- ▶ Example tools:
 - ▶ `pg_dump` for single database.
 - ▶ `pg_dumpall` for full cluster backup.

pg_dump

```
postgres@ip-172-31-27-8:~$ pg_dump --help
pg_dump dumps a database as a text file or to other formats.

Usage:
  pg_dump [OPTION]... [DBNAME]

General options:
  -f, --file=FILENAME          output file or directory name
  -F, --format=cldltlp         output file format (custom, directory, tar,
                                plain text (default))
  -j, --jobs=NUM               use this many parallel jobs to dump
  -v, --verbose                 verbose mode
  -V, --version                 output version information, then exit
  -Z, --compress=0-9           compression level for compressed formats
  --lock-wait-timeout=TIMEOUT  fail after waiting TIMEOUT for a table lock
  --no-sync                     do not wait for changes to be written safely to disk
  -?, --help                   show this help, then exit

Options controlling the output content:
  -a, --data-only              dump only the data, not the schema
  -b, --blobs                  include large objects in dump
  -B, --no-blobs               exclude large objects in dump
  -c, --clean                  clean (drop) database objects before recreating
```

Pros and Cons of Logical Backup

Pros:

- ▶ Can back up specific parts of the database (not just the entire cluster).
- ▶ Simple and human-readable format.
- ▶ Backups are stored as plain text/SQL files, making them portable across different operating systems.
- ▶ Faster than physical backup when only a single DB object is required.

Cons:

- ▶ Configuration files are not included (must be backed up separately).
- ▶ Point-in-Time Recovery (PITR) is not supported.

Plan

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Types of Backups

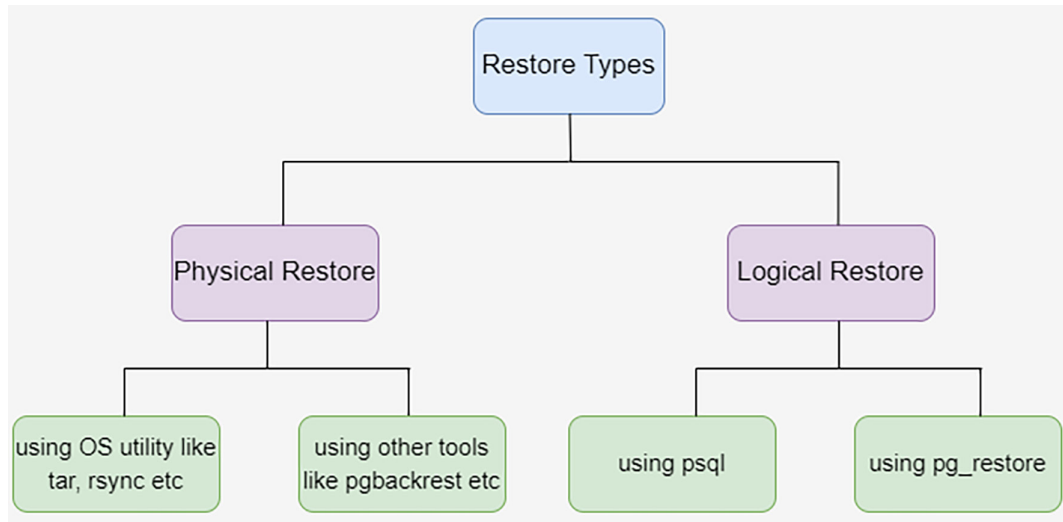
Physical

Logical

Types of Restore

Advanced Backup Tools

Types of Restore



Restore Methods

- ▶ `psql`: Restores backups in plain SQL format.
- ▶ `pg_restore`: Restores backups from compressed/custom format.

pg_restore

```
postgres@ip-172-31-27-8:~$ pg_restore --help
pg_restore restores a PostgreSQL database from an archive created by pg_dump.
```

Usage:

```
pg_restore [OPTION]... [FILE]
```

General options:

-d, --dbname=NAME	connect to database name
-f, --file=FILENAME	output file name (- for stdout)
-F, --format=cldlt	backup file format (should be automatic)
-l, --list	print summarized TOC of the archive
-v, --verbose	verbose mode
-V, --version	output version information, then exit
?, --help	show this help, then exit

Options controlling the restore:

-a, --data-only	restore only the data, no schema
-c, --clean	clean (drop) database objects before recreating
-C, --create	create the target database
-e, --exit-on-error	exit on error, default is to continue
-I, --index=NAME	restore named index

Logical Backup and Restore example

Description	Command
To dump a database called mydb into a SQL-script file:	<code>pg_dump mydb > "/opt/mydb_dump.sql"</code>
To reload such a script into a (freshly created) database named newdb:	<code>psql -d newdb -f "/opt/mydb_dump.sql"</code>
To dump a database into a custom-format archive file:	<code>pg_dump -Fc mydb > "/opt/mydb_dump.dump"</code>
To reload an archive file into a (freshly created) database named newdb:	<code>pg_restore -d newdb "/opt/mydb_dump.dump"</code>
To dump a single table named mytab	<code>pg_dump -t mytab mydb > "/opt/mydb_dump.sql"</code>
To dump the database with the log file using user test	<code>pg_dump -U test -v -f "/opt/mydb_dump.sql" mydb 2>> "/opt/mydb_log.log"</code> password for user test :
To dump only structure without data	<code>pg_dump -sU test -v -f "/opt/mydb_dump.sql" mydb 2>> "/opt/mydb_log.log"</code> password for user test :
To take insert scripts of particular table	<code>pg_dump --column-inserts -a -t mytab -U test mydb > "/opt/mytab_inserts.sql"</code>
To dump only specific tables with data	<code>pg_dump -U test -n schema1 -t BCL_* -f "/opt/BCL_TABLES.sql" mydb</code>

Useful Backup and Restore Tools

Community-developed tools:

- ▶ Support full, incremental, and differential backups.
- ▶ Provide backup compression and checksum.
- ▶ Allow multiple repositories: on-premise or cloud.
- ▶ Enable parallelism for faster operations.
- ▶ Maintain a backup catalog.
- ▶ Offer alerting and notification features.

We will next review some of the most popular open-source tools, their options, and how to perform a backup and restore.

Plan

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Types of Backups

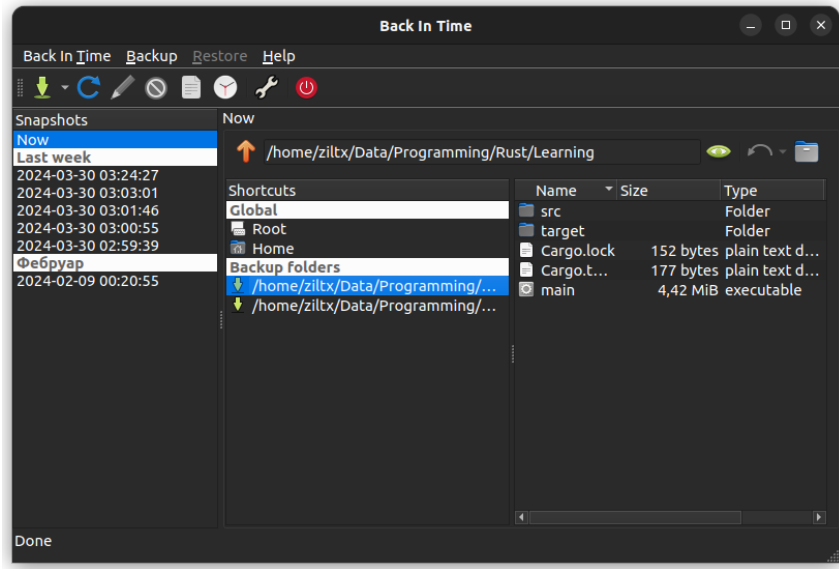
Physical

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Types of Restore

Advanced Backup Tools

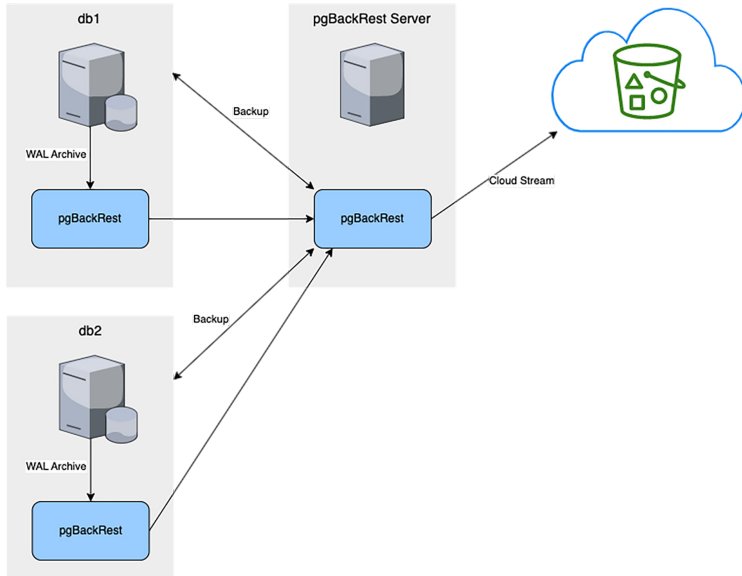




pgBackRest

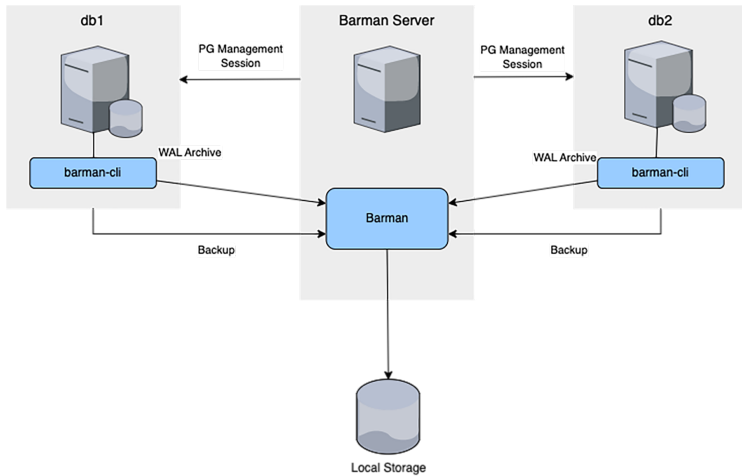
- ▶ Supports full, incremental, and differential backups.
- ▶ Uses parallelism for faster operations.
- ▶ Supports cloud storage.

pgBackRest



- ▶ Used for remote backups.
- ▶ Provides incremental backup and deduplication.
- ▶ Maintains a backup catalog.

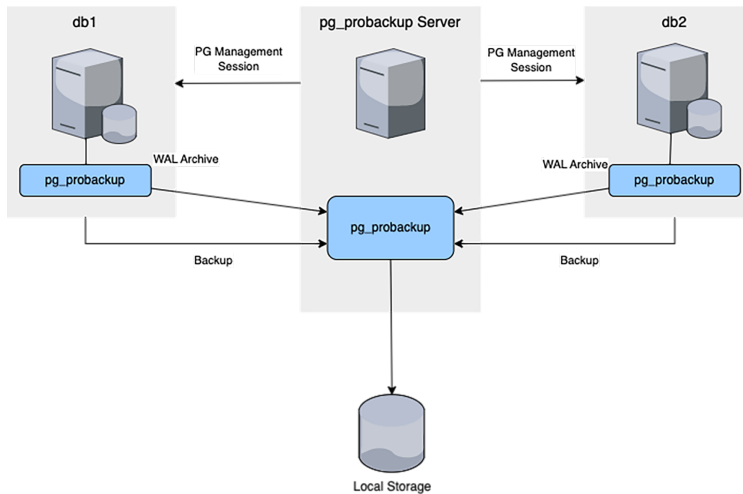
Barman



pg_probackup

- ▶ Provides incremental restore to save time.
- ▶ Supports backup merging and deduplication.
- ▶ Offers parallelism and remote operations.

pg_probackup





- 5 **Regular backups** are essential for ensuring high availability, enabling point-in-time recovery, and recovering from accidental changes.

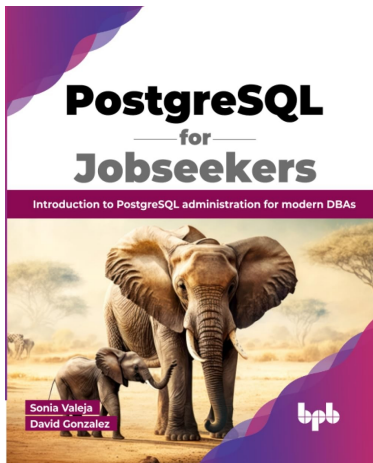
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- 2 **Point-in-Time Recovery (PITR)** requires WAL archiving and is only supported with physical backups, not logical ones.
- 1 **Community tools** such as `pgBackRest`, `Barman`, and `pg-probackup` provide advanced features like incremental backups, compression, cloud storage, and catalogs.

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